The numbers 0 and 1 – Answer key

Exercise 39

1 True or false?

a) **False**: 8 × 0 = 0

b) **True**

c) **True**

d) **False**: 9 × 1 = 9

e) **False**: we cannot divide by 0

f) **True**

2 Calculate

a) (−12) × 0 = **0**

b) (−15) × 1 = **−15**

c) 0 ÷ (−4) = **0**

d) 9 + (−9) = **0**

e) (−6) ÷ (−6) = **1**

f) 23 × 1 × 0 = **0**

3 Find the missing number

a) 5 × ☐ = 0 ☐ = **0**

b) ☐ + (−3) = 0 ☐ = **3**

c) ☐ × 1 = −8 ☐ = **−8**

d) 0 ÷ ☐ = 0 ☐ cannot be **0**

> 3d) Any number except 0 works (0 ÷ 7 = 0, 0 ÷ (−2) = 0), but 0 ÷ 0 is not possible because we cannot divide by 0.

4 Using 0 and 1 to help

a) 358 × 0 + 12 = 0 + 12 = **12**

b) 49 × 1 − 49 = 49 − 49 = **0**

c) 0 × 15 − 3 = 0 − 3 = **−3**

5 Maya says: "6 ÷ 0 = 0"

Any good answer: 6 ÷ 0 asks "which number times 0 gives 6?" But any number times 0 is 0, never 6, so no number works. We cannot divide by 0. Maya may be thinking of 0 ÷ 6 = 0, which is fine (0 × 6 = 0).

CHALLENGE

5 ÷ 0 asks "? × 0 = 5". Anything times 0 is 0, so no number can be the answer. The calculator shows an error because there is no answer to show.